

almost zero, it may not be necessary to price them. But its supply does cost, say in terms of high fixed cost, it is suggested that it should be public funded. Then, there exists problem of free-ridership as true preferences for such goods may not get revealed.

Public goods are goods which are non-rival in consumption and non-excludable from consumption whereas private goods which are both rival and excludable. However, it is not true that all goods in existence can be categorised in these two mutually exclusive groups. Goods may be non-rival but still excludable. Few more students may be accommodated in the class as marginal cost of teaching them is zero and consumption is non-rival up to a limit when the class size becomes too large for effective learning to take place. Such goods are often called club goods. There are goods, such as fishing in international water, where fishers are non-excludable, but extraction of fish is rival: more for me and less for you or more today and less tomorrow. So may be the case with oil extraction from two nearby wells with common stock underground. Such goods are often referred to as commons or common goods. Club goods and common goods are also referred to as quasi-public goods or semi-public goods.

For pure private goods, equilibrium condition $MSC = MB_A = MB_B$ where A and B are two consumers, decides the quantum of good that would be produced while in case of pure public goods, equilibrium condition $MSC = MB_A + MB_B$ should decide that quantum. In the case of a private good, marginal benefit curves are horizontally added while in the case of public good, they have to be vertically added. Needless to say, market may not produce any such good. State is therefore called for, for production of public goods.

We may further note that (1) public goods may be national or local or global depending upon their area of outreach and (2) there also exist public bads (global, national and local), like variety of pollutions, which have to be reduced or removed by the State.

6.4.3 Externalities

Most activities related to production of goods, and some involving consumption of goods, also produce bads, which impact others, called third parties– as they are neither producer (first party) nor consumers (second party) who benefit from production of the good. This phenomenon is known as externalities. Sugar or textile or leather factories produce effluents which find way into local water streams impacting those living nearby or depending on them for water. People living in the vicinity suffer from the polluted streams in terms of health hazards and incur expenditure on protection or medical treatment. Some of the production activities do produce benefits which accrue to those who do not pay for those benefits. A classic example is beekeeping. While collecting nectar bees pollinate the trees in the orchard. Free software or basic education are treated in the same category. Thus, we see externalities are negative or positive. The short point is that social cost or social benefit diverge from their private counterpart. Third party cost/benefits are also known as external cost/external benefit (EC/EB). Generally, marginal equilibrium conditions are written as:

For Negative Externality: $MSC = MC (MPEC) + MEC$ while $MR = MSB$

For Positive Externality: $MSB = MR (MPEB) + MEB$ while $MC = MSC$

where Marginal Private Economic Cost (MPEC) represents marginal cost and Marginal Private Economic Benefit (MPEB), marginal revenue. MEC and MEB are Marginal External Cost borne and Marginal External Benefit enjoyed by a third party. Since private parties tend to equate marginal (private) cost with marginal (private) benefit, quantum produced may not be socially efficient. While focusing on negative externalities, it was argued by Pigou to adopt 'polluter pays' principle. The suggestion is externalities should be internalised. Since then, many other suggestions have been offered. Allocation of tradable pollution permits is one. Subsidisation sought by private parties to adopt technologies to help reduce pollution is another. Coase had argued for assignment of property rights and private negotiation, but negotiation need to be enforced by State. So, State intervention is needed in such cases as well.

6.4.4 Monopoly Power

When there is only one producer, monopoly is obvious but monopoly power in terms of fixing price is not that obvious. A monopolist dealing in a commodity with an inelastic demand has higher monopoly power than one who is dealing in a commodity with an elastic demand. Elasticity of the demand curve faced by a firm depends on the number of firms in the market as also on substitutes (implying cross elasticity). Oligopolistic and monopolistically competitive firms have thus also some monopoly power. Thus, there is a degree of monopoly with firms not in a fully competitive market. Since for a firm in a fully competitive market, price is equal to marginal cost ($P = MC$), the divergence between the two, that is excess of P over MC , will determine the monopoly power. Lerner devised an index with formula as

$$\text{Degree of Monopoly Power} = (P - MC)/P,$$

which is zero when $P = MC$ and 1 when marginal cost is zero. Since MC is equal to MR in equilibrium and $MR = P(1 - (1/e))$ where e is the measure of elasticity, the degree of monopoly can be shown to be equal to $1/e$. Monopoly power is inverse of the elasticity of demand.

There may be good reason to permit monopolies in certain cases. Like in natural monopolies, it benefits the society to harness potential of a technology in terms of economies of scale. However, if producers are few or if a producer controls a good proportion of market, they enjoy market power, which manifests in the form of goods offered at a higher prices. In both cases, State is invited to intervene by regulating prices in addition to quantity and quality so that social outcomes are optimum. Monopoly restricting laws have been enacted in most countries. Competition Commissions are also instituted by the governments in many countries, including in India, to oversee that market power or market dominance is not misused to the disadvantage of consumers.

6.4.5 Information Asymmetries

One of the basic assumptions behind perfect competition is prevalence of full and complete knowledge among buyers and sellers of goods, which is hardly

ever the case in reality. With growing economic complexities, though knowledge has increased, so has ignorance or lack of adequate information to make good economic decisions. There is another aspect, information is often asymmetric between the two parties that may be involved in pursuing a transaction. State tries to reduce this asymmetry. There are acts and institutions setting quality standards to take care of— safety in case food articles, quality in case of metals, control of pollution, etc. Samuelson quotes the case of Eben Byers who was a steel mogul in the US. He took drug Radithor sold as an aphrodisiac and cure-all to relieve himself from his ailments. Byers died from its consumption. Later on, it was discovered that Radithor was distilled water laced with radium. State has a role in addressing such informational deficiencies.

6.4.6 Merit and Demerit Goods

There are often goods demanded by some people that may not be considered desirable by the society. State as an important pillar of society is expected to encourage consumption of merit goods. For example, children may not value education and its demand may be low. Competitive market may produce very little education to meet the needs of everyone. State encourages its consumption by subsidising it or making it free of tuition fee. Inoculation, immunisation, vaccination, sponsorship of cultural festivals may all fall in that category. Similarly, there may be goods such as intoxicating drugs tobacco, cigarettes or alcohol, the consumption of which a State may want to discourage. State could discourage consumption through several ways including imposing heavy taxes, restricting or banning the production. Some States in India have prohibited the consumption of alcohol or tobacco. At the moment, Gujarat and Bihar have banned production, imports and consumption of alcohol, while Haryana has lifted the ban imposed earlier.

Check Your Progress 3

- 1) What are the reasons for State interventions in specific markets?
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- 2) Explain the rationale of State intervention in the market on account of externalities in production and consumption.
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